



WINTRIX ACADEMY PRESENTS

DCET Previous Year Question Paper & Solutions

2024

DIPLOMA COMMON ENTRANCE TEST

GUIDANCE • STRATEGY • SUCCESS

100 Questions • Detailed Solutions • Subject-Wise Organisation

FIRST EDITION — 2026



DCET 2024 Previous Year Question Paper & Solutions

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Disclaimer: This book contains previous year question papers from the DCET examination conducted by the Karnataka Examination Authority (KEA). The solutions and explanations are prepared by Wintrix Academy for educational purposes.

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About Wintrix Academy

Wintrix Academy is a premier DCET mentorship platform founded with a singular mission: to empower Diploma students across Karnataka with the guidance, strategy, and resources they need to secure top ranks in the Diploma Common Entrance Test.



Mohammed Adnan

FOUNDER & CHIEF MENTOR

Mohammed Adnan secured All-Karnataka DCET Rank 48 and 2nd Rank in Hassan District, earning admission to R.V. College of Engineering (RVCE) for Computer Science Engineering with a current CGPA of 9.19. Placed at Red Hat.

48

DCET RANK

9.8

DIPLOMA
CGPA

9.19

RVCE CGPA

50+

STUDENTS



Jnanesh Gowda

DIRECTOR & BRAND ARCHITECT

Turning confusion into clarity and effort into results. Jnanesh brings new-age learning strategies and results-engineered systems to ensure every student's success is built by design.

Preface

Dear DCET Aspirant,

This book contains the complete DCET 2024 Question Paper along with detailed solutions for all 100 questions. Every solution has been prepared to help you understand the reasoning behind each answer.

The questions are organised subject-wise across five chapters — Engineering Mathematics, Project Management Skills, IT Skills, Statistics & Analytics, and Fundamentals of Electrical & Electronics Engineering — each carrying 20 marks.

Practising previous year papers is one of the most effective strategies for DCET preparation. I encourage you to attempt each question on your own before looking at the solution.

Wishing you all the very best for your DCET journey.

Mohammed Adnan

Founder, Wintrix Academy
DCET Rank 48 | RVCE CSE

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100 QUESTIONS • 5 SUBJECTS • 20 MARKS EACH • 100 MARKS TOTAL

01

CHAPTER ONE

Engineering Mathematics

20 Marks

Questions 1 – 20

Q.01

MATRICES

If $A = \begin{bmatrix} 3 & -2 \\ 4 & 1 \end{bmatrix}$ and $A - B = \begin{bmatrix} 5 & -3 \\ 4 & 6 \end{bmatrix}$, then matrix B is

a) $\begin{bmatrix} -3 & 2 \\ -4 & -1 \end{bmatrix}$

b) $\begin{bmatrix} -5 & 3 \\ -4 & -6 \end{bmatrix}$

c) $\begin{bmatrix} -2 & 1 \\ 0 & -5 \end{bmatrix}$

d) $\begin{bmatrix} 2 & -1 \\ 0 & 5 \end{bmatrix}$

✓ ANSWER: (C)

$$B = A - (A - B) = \begin{bmatrix} 3 & -2 \\ 4 & 1 \end{bmatrix} - \begin{bmatrix} 5 & -3 \\ 4 & 6 \end{bmatrix}$$

$$B = \begin{bmatrix} 3-5 & -2+3 \\ 4-4 & 1-6 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 0 & -5 \end{bmatrix}$$

Q.02

CRAMER'S RULE

If $\begin{bmatrix} 3 & -4 \\ 9 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \end{bmatrix}$, then the values of x and y are

a) $\left(\frac{2}{3}, -2 \right)$

b) $\left(2, \frac{2}{3} \right)$

c) $\left(-2, \frac{2}{3} \right)$

d) $\left(-\frac{2}{3}, 2 \right)$

✓ ANSWER: (A)

$$3x - 4y = 10 \dots (i)$$

$$9x + 2y = 2 \dots (ii)$$

$$\text{Multiply (i) by 2: } 6x - 8y = 20$$

$$\text{Multiply (ii) by 4: } 36x + 8y = 8$$

$$\text{Adding: } 42x = 28 \rightarrow x = \frac{2}{3}$$

$$\text{Substituting: } 3\left(\frac{2}{3}\right) - 4y = 10 \rightarrow y = -2$$

If $A = \begin{bmatrix} -2 & 6 \\ -2 & 3 \end{bmatrix}$, then A^{-1} is

a) $\frac{1}{18} \begin{bmatrix} 3 & -6 \\ 2 & -2 \end{bmatrix}$

b) $\frac{1}{6} \begin{bmatrix} 3 & -6 \\ 2 & -2 \end{bmatrix}$

c) $\begin{bmatrix} 3 & -6 \\ 2 & -2 \end{bmatrix}$

d) $\frac{1}{6} \begin{bmatrix} -2 & 6 \\ -2 & 3 \end{bmatrix}$

✓ ANSWER: (B)

$$\begin{vmatrix} -2 & 6 \\ -2 & 3 \end{vmatrix} = (-2)(3) - (6)(-2) = -6 + 12 = 6$$

$$\text{adj}(A) = \begin{bmatrix} 3 & -6 \\ 2 & -2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj}(A) = \frac{1}{6} \begin{bmatrix} 3 & -6 \\ 2 & -2 \end{bmatrix}$$

If $x + y = 3$ and $2x + 3y = 8$ are the linear equations, then x and y are

a) $(1, -2)$

b) $(1, 2)$

c) $(-1, -2)$

d) $(-1, 2)$

✓ ANSWER: (B)

$$x + y = 3 \rightarrow \text{multiply by } 2: 2x + 2y = 6$$

$$2x + 3y = 8$$

$$\text{Subtracting: } -y = -2 \rightarrow y = 2$$

$$x = 3 - 2 = 1 \therefore (x, y) = (1, 2)$$

Q.05

STRAIGHT LINES

If the straight line is $2y = 3x + 4$, then the y-intercept is

a) 4

b) 2

c) $-\frac{4}{3}$

d) -2

✓ ANSWER: (B)

$$2y = 3x + 4 \rightarrow 3x - 2y + 4 = 0$$

$$\text{y-intercept} = \frac{-c}{b} = \frac{-4}{-2} = 2$$

Q.06

STRAIGHT LINES

The acute angle of inclination made by the line $3x - 3y - 2 = 0$ with positive x-axis is

a) 135°

b) 30°

c) 45°

d) 0°

✓ ANSWER: (C)

$$\text{Slope } m = \frac{-a}{b} = \frac{-3}{-3} = 1$$

$$\tan \theta = 1 \rightarrow \theta = 45^\circ$$

Q.07

STRAIGHT LINES

The slope of a straight line passing through the points (1, 2) and (5, -3) is

a) $\frac{5}{4}$

b) $\frac{1}{6}$

c) $-\frac{1}{4}$

d) $-\frac{5}{4}$

✓ ANSWER: (D)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-3 - 2}{5 - 1} = -\frac{5}{4}$$

If the slope of the straight line is $-\frac{3}{2}$ and y-intercept is $\frac{5}{2}$, then the equation of the straight line is

a) $3x + 2y - 5 = 0$

b) $3x - 2y - 5 = 0$

c) $3x + 2y + 5 = 0$

d) $3x - 2y + 5 = 0$

✓ ANSWER: (A)

$$y = mx + c = \left(-\frac{3}{2}\right)x + \frac{5}{2}$$

$$2y = -3x + 5 \rightarrow 3x + 2y - 5 = 0$$

The line perpendicular to the straight line $5x - y + 1 = 0$ and passing through the point (2, 3) is

a) $x - 5y - 17 = 0$

b) $x + 5y - 13 = 0$

c) $x + 5y - 17 = 0$

d) $x + 5y + 13 = 0$

✓ ANSWER: (C)

Slope of $5x - y + 1 = 0$ is $m = 5$

Slope of perpendicular = $-\frac{1}{5}$

$$y - 3 = \left(-\frac{1}{5}\right)(x - 2)$$

$$5y - 15 = -x + 2 \rightarrow x + 5y - 17 = 0$$

The value of $\cot 660^\circ$ is

a) $-\frac{1}{\sqrt{3}}$

b) $\sqrt{3}$

c) $\frac{1}{\sqrt{3}}$

d) $-\sqrt{3}$

✓ ANSWER: (A)

$$\begin{aligned}\cot 660^\circ &= \cot(360^\circ + 300^\circ) = \cot 300^\circ \\ &= \cot(270^\circ + 30^\circ) = -\tan 30^\circ = -\frac{1}{\sqrt{3}}\end{aligned}$$

The value of $\cos 75^\circ$ is

a) $\frac{\sqrt{3} + 1}{2\sqrt{2}}$

b) $\frac{\sqrt{3} + 1}{\sqrt{3} - 1}$

c) $\frac{\sqrt{3} - 1}{2\sqrt{2}}$

d) $\frac{\sqrt{3} - 1}{\sqrt{3} + 1}$

✓ ANSWER: (C)

$$\begin{aligned}\cos 75^\circ &= \cos(45^\circ + 30^\circ) \\ &= \cos 45^\circ \cdot \cos 30^\circ - \sin 45^\circ \cdot \sin 30^\circ \\ &= \left(\frac{1}{\sqrt{2}} \right) \left(\frac{\sqrt{3}}{2} \right) - \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{2} \right) \\ &= \frac{\sqrt{3} - 1}{2\sqrt{2}}\end{aligned}$$

If a ladder is inclined to the wall with base angle 30° , then the angle in radians is

a) $\frac{\pi}{3}$

b) $\frac{\pi}{4}$

c) $\frac{\pi}{2}$

d) $\frac{\pi}{6}$

✓ ANSWER: (D)

$$\text{Angle in radian} = \left(\frac{\pi}{180^\circ} \right) \times 30^\circ = \frac{\pi}{6} \text{ radian}$$

If $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, then $\cos(A - B)$ is

a) $-\frac{33}{65}$

b) $\frac{63}{65}$

c) $-\frac{63}{65}$

d) $\frac{33}{65}$

✓ ANSWER: (B)

$$\begin{aligned}\sin A &= \frac{3}{5} \rightarrow \cos A = \frac{4}{5} \text{ (using 3-4-5 triplet)} \\ \cos B &= \frac{12}{13} \rightarrow \sin B = \frac{5}{13} \text{ (using 5-12-13 triplet)} \\ \cos(A-B) &= \cos A \cdot \cos B + \sin A \cdot \sin B \\ &= \left(\frac{4}{5} \right) \left(\frac{12}{13} \right) + \left(\frac{3}{5} \right) \left(\frac{5}{13} \right) \\ &= \frac{48}{65} + \frac{15}{65} = \frac{63}{65}\end{aligned}$$

Q.14

DIFFERENTIAL CALCULUS

If $y = 2 \sin x - 4 \cot x + e^{-x} + 7$, then dy/dx is

- a) $-2 \cos x - 4 \operatorname{cosec}^2 x - e^{-x} + 7x$ **b) $2 \cos x + 4 \operatorname{cosec}^2 x - e^{-x}$**
- c) $2 \cos x + 4 \operatorname{cosec}^2 x + e^x + 7$ d) $2 \sin x - 4 \operatorname{cosec}^2 x + e^x$

✓ ANSWER: (B)

$$\begin{aligned} dy/dx &= 2 \cos x - 4(-\operatorname{cosec}^2 x) - e^{-x} \\ &= 2 \cos x + 4 \operatorname{cosec}^2 x - e^{-x} \end{aligned}$$

Q.15

DIFFERENTIAL CALCULUS

If $y = (x + \frac{1}{x})^2$, then dy/dx is

- a) $x^2 + \frac{1}{x^2} + 2$ b) $2x + \frac{1}{x^2}$
- c) $2x - \frac{2}{x^3}$** d) $2x + \frac{1}{x}$

✓ ANSWER: (C)

$$\begin{aligned} y &= (x + \frac{1}{x})^2 = x^2 + \frac{1}{x^2} + 2 \\ dy/dx &= 2x - \frac{2}{x^3} \end{aligned}$$

Q.16

DIFFERENTIAL CALCULUS

If $y = e^{3x} + e^{7x}$, then d^2y/dx^2 at $x = 0$ is

- a) 10 **b) 58**
- c) 1 d) 2

✓ ANSWER: (B)

$$\begin{aligned} dy/dx &= 3e^{3x} + 7e^{7x} \\ d^2y/dx^2 &= 9e^{3x} + 49e^{7x} \\ \text{At } x = 0: &9(1) + 49(1) = 58 \end{aligned}$$

If $y = \log(\sec x + \tan x)$, then dy/dx is

a) $\sec x + \tan x$

b) $\frac{1}{\sec x + \tan x}$

c) $\sec x$

d) $\sec^2 x$

✓ ANSWER: (C)

$$\begin{aligned} dy/dx &= [\sec x \tan x + \sec^2 x] / (\sec x + \tan x) \\ &= \sec x (\tan x + \sec x) / (\sec x + \tan x) \\ &= \sec x \end{aligned}$$

$\int (3x^2 + 2x + 2)^{3/2} (3x + 1) dx$ is

a) $\frac{3x+5}{5} + C$

b) $\frac{3x-5}{5} + C$

c) $\left(\frac{3x^2 - 2x - 2}{5} \right)^{5/2} + C$

d) $\left(3x^2 + \frac{2x}{5} + 2 \right)^{5/2} + C$

✓ ANSWER: (D)

$$\begin{aligned} \text{Let } f(x) &= 3x^2 + 2x + 2, \quad f'(x) = 6x + 2 = 2(3x+1) \\ \int f(x)^{3/2} (3x+1) dx &= \frac{1}{2} \int f(x)^{3/2} \cdot f'(x) dx \\ &= \frac{1}{2} \cdot \frac{f(x)^{5/2}}{5/2} + C \\ &= \frac{(3x^2 + 2x + 2)^{5/2}}{5} + C \end{aligned}$$

Q.19

INTEGRAL CALCULUS

The integral of $x^3 + \frac{1}{x} + e^x$ is

a) $\frac{x^4}{4} + \log x - e^x + C$

b) $\frac{x^4}{4} + \log x + e^x + C$

c) $3x^2 - \frac{1}{x^2} + e^x + C$

d) $3x^2 - \frac{1}{x^2} - e^x + C$

✓ ANSWER: (B)

$$\int (x^3 + \frac{1}{x} + e^x) dx = \frac{x^4}{4} + \log x + e^x + C$$

Q.20

INTEGRAL CALCULUS

$\int x \sin x \, dx =$

a) $x \cos x - \sin x + C$

b) $-x \cos x + \sin x + C$

c) $x \cos x + \sin x + C$

d) $-x \cos x - \sin x + C$

✓ ANSWER: (B)

Using integration by parts: $\int u \, dv = uv - \int v \, du$
 $u = x, \, dv = \sin x \, dx$
 $= x(-\cos x) - \int (-\cos x) dx$
 $= -x \cos x + \sin x + C$

02

CHAPTER TWO

Project Management Skills

20 Marks

Questions 21 – 40

Q.21

PM BASICS

Which of the following are not obstacles in project management?

- a) Project complexities
- b) Project risks
- c) Changes in technology
- d) Early identification of problems

✓ ANSWER: (D)

Early identification of problems is a benefit of project management, not an obstacle.

Q.22

CONSULTANTS

_____ provides guidance as well as direction to the project and he/she is not a part of the project team.

- a) CEO
- b) Project Manager
- c) Team Leader
- d) Project Consultant

✓ ANSWER: (D)

Project Consultant guides the project but is not a part of the project team.

Q.23

PROJECT TYPES

Repair of dam in case of damage due to natural calamities is an example of

- a) Normal project
- b) Crash project
- c) Risky project
- d) Disaster project

✓ ANSWER: (D)

A project performed due to the effect of natural calamities is known as a **disaster project**.

_____ refers to a detailed description of the expected outcome of a project.

- a) Project scope
- b) Project operation
- c) Project objective**
- d) Project process

✓ ANSWER: (C)

Project objective illustrates the tasks you execute to achieve the result.

The process of dividing the project into number of tasks in order to carry out effective project administration is:

- a) PPM
- b) WBS**
- c) PEP
- d) Project diary

✓ ANSWER: (B)

Work Breakdown Structure (WBS) is the process of dividing project work into tasks.

The project _____ relieves a project team from most regular work such as planning, tracking and reporting responsibility.

- a) Manager
- b) CEO
- c) Administration**
- d) Leader

✓ ANSWER: (C)

Project **administration** relieves a project team from regular work.

The use of authority to channelize the activities of the project on desired lines is referred to as

- a) Project direction**
- b) Project coordination
- c) Project communication
- d) Project execution

✓ ANSWER: (A)

Project direction channelizes the activities of the project on desired lines.

Q.28**WBS**

_____ illustrates total project work into divisions of major work packages.

- a) PERT
- b) CPM
- c) WBS**
- d) PEP

✓ ANSWER: (C)

WBS divides the work into major packages.

Q.29**OVERRUNS**

_____ is a condition wherein a construction or manufacturing project does not complete well within its prescribed time schedule.

- a) Time analysis
- b) Time estimate
- c) Time overrun**
- d) Cost overrun

✓ ANSWER: (C)

Time overrun is when a project cannot be completed within its prescribed time schedule.

Q.30**RISK**

A project _____ creates damages or adverse effects to either project activities or project deliverables.

- a) Failure
- b) Undertaken
- c) Risk**
- d) Analysis

✓ ANSWER: (C)

Damages or uncertainties affecting a project is known as **risk**.

Q.31**RISK ANALYSIS**

Identify the correct risk analysis which has typically one variable changed at a time

- a) Sensitivity Analysis**
- b) Scenario Analysis
- c) Best and Worst Case Analysis
- d) Simulation Analysis

✓ ANSWER: (A)

Sensitivity Analysis is a type of risk analysis done by changing one variable at a time.

Q.32**RISK**

Risk usually _____ as the project progresses.

- a) Increases
- b) Reduces
- c) Remains same
- d) Becomes negligible

✓ ANSWER: (B)

As a project progresses towards its goal, risk usually **reduces**.

Q.33**SMART**

In 'SMART', the letter R stands for _____.

- a) Realistic
- b) Reliable
- c) Reasonable
- d) Relation

✓ ANSWER: (A)

SMART: S-Specific, M-Measurable, A-Achievable, R-**Realistic**, T-Timely

Q.34**EVALUATION**

_____ is a step-by-step process of collecting, recording and organising information about project results.

- a) Project planning
- b) Project evaluation
- c) Project scheduling
- d) Project monitoring

✓ ANSWER: (B)

Project evaluation is a process where information is collected, recorded, and organised to check project results.

Q.35**POLICIES**

Which one of the following is not a policy of a project?

- a) Extent of work given to outside contractors
- b) Number of contractors to be employed
- c) Monitoring the progress of project
- d) Terms of contract

✓ ANSWER: (C)

Monitoring the progress is a phase of project life cycle, not a policy.

Q.36**SCHEDULING**

In scheduling, 'EST' stands for:

- a) Easiest Start Time
- b) Earliest Start Time**
- c) Earliest Schedule Time
- d) Earliest Schedule Technique

✓ ANSWER: (B)

EST stands for **Earliest Start Time**.

Q.37**AUDITOR**

_____ is an expert in measuring, confirming, investigating and reporting status of a project.

- a) Project Guide
- b) Project Reviewer
- c) Project Auditor**
- d) Project Advisor

✓ ANSWER: (C)

Project Auditor is the one who reports the status of a project.

Q.38**TIME ESTIMATES**

_____ is the time estimate if there are no hurdles in an activity.

- a) Pessimistic time
- b) Optimistic time**
- c) Most likely time
- d) Expected estimated time

✓ ANSWER: (B)

Optimistic time is where project time is scheduled based on quickest completion without any risk.

Q.39**LIFE CYCLE**

_____ is the last phase of the project life cycle which reports overall achievements of a project in terms of defined performance measures.

- a) Project closure**
- b) Project initiation
- c) Project execution
- d) Project planning

✓ ANSWER: (A)

Project closure is the final phase of the project life cycle.

The objective of network analysis in a project is to

- a) Minimise total project duration
- b) Minimise total project cost
- c) Minimise production delays
- d) Minimise manpower requirements

✓ ANSWER: (A)

Minimise total project duration is the objective of network analysis.

03

CHAPTER THREE

IT Skills

20 Marks

Questions 41 – 60

Q.41

PSEUDOCODE

What is the output of the following?

a = 10

if ($a > 1$) then

```
print "TWENTY"
```

else

```
print "TEN"
```

- a) 10 b) TEN
- c) TWENTY d) No Output

✓ ANSWER: (C)

Since $a = 10$, and $10 > 1$ is true, the output is **TWENTY**.

Q.42

FLOWCHARTS

The symbol used to connect two parts of a big flowchart is

- a) Oval b) Rectangle
c) Diamond **d) Circle**

✓ ANSWER: (D)

In flowcharts, **circles** are used as connector symbols to show the continuation of flow across multiple pages.

Q.43

FLOWCHARTS

Which symbol is used to represent start or stop in a flowchart?

- a) Rectangle b) Diamond
- c) Oval d) Circle

✓ ANSWER: (C)

The **oval** symbol represents the start and end points of a process.

Q.44

CSS

What does CSS stand for?

- a) Colourful Style Sheet
- b) Cascading Style Sheet**
- c) Creative Style Sheet
- d) Computer Style Sheet

✓ ANSWER: (B)

CSS stands for **Cascading Style Sheet**.

Q.45

WEB

Which of the following is not a web browser?

- a) Google Chrome
- b) Netscape Navigator
- c) Safari
- d) Apache**

✓ ANSWER: (D)

Apache is a web server software, not a web browser.

Q.46

CSS

What is the correct syntax to change the background colour with CSS?

- a) background-color = "purple";
- b) background-color : purple;**
- c) background-color = orange
- d) change-background-color : purple

✓ ANSWER: (B)

In CSS, properties are separated from values by a colon. **background-color: purple;**

Q.47

HTML

Which HTML tag gives a scrolling effect of a given text?

- a) <div>
- b) <scroll>
- c) <marquee>
- d) <effect>

✓ ANSWER: (C)

The **<marquee>** tag is used to create scrolling text in HTML.

Q.48

ERP

ERP package will handle _____ business functionality/functionalities.

- a) One
- b) Two
- c) Three
- d) Multiple

✓ ANSWER: (D)

ERP integrates all functional units — it handles **multiple** business functionalities.

Q.49

ORGANIZATION

Which one is a type of organizational structure?

- a) Responsibility structure
- b) Behavioural structure
- c) Functional structure
- d) Relational structure

✓ ANSWER: (C)

Functional structure is a type of organisational structure based on functional units.

Q.50

USE CASE

The _____ describes how an actor uses a system to accomplish a particular goal for describing the activities.

- a) Process case
- b) Work case
- c) Use case
- d) Sub process

✓ ANSWER: (C)

Use case describes how an employee uses a system.

Which of the following is not a characteristic of First-generation ERP system?

- a) They were based on mainframe computer
- b) They focused on accounting and finance
- c) They used centralized database
- d) They supported real-time data processing**

✓ ANSWER: (D)

First-generation ERP systems were standalone applications for specific functions. **Real-time data processing** was not a characteristic.

Which cloud service model enables users to develop, run, manage application without worrying about underlying infrastructure?

- a) IaaS
- b) PaaS**
- c) SaaS
- d) NaaS

✓ ANSWER: (B)

PaaS (Platform as a Service) provides a platform for developing and managing applications without infrastructure concerns.

Which of the following is not a Google application?

- a) Google Docs
- b) Google Slides
- c) Google Sheets
- d) Google Word**

✓ ANSWER: (D)

Google Word does not exist. Google's word processing application is called Google Docs.

IoT is a _____.

- a) Network of sensors
- b) Network of objects in a ring structure
- c) Network of virtual objects
- d) Network of physical objects embedded with sensors**

✓ ANSWER: (D)

IoT refers to the network of physical objects embedded with sensors, software, and other technologies for connecting and exchanging data.

_____ is a file storage and synchronization service developed by Google which provides complete office tools with cloud storage.

- a) Google Colab
- b) Google Drive**
- c) Google Meet
- d) AWS Cloud

✓ ANSWER: (B)

Google Drive is a file storage and synchronization service with cloud storage and office tools integration.

Which of the following is not a security issue in a personal computer?

- a) Ransomware Attack
- b) Data Breach
- c) Code Injection
- d) IoT**

✓ ANSWER: (D)

IoT is a technology concept, not a security issue in personal computers.

Q.57**FIREWALL**

Which of the following is not a firewall?

- a) Packet filtering
- b) Stateful inspection
- c) Panda**
- d) Circuit Level Gateway

✓ ANSWER: (C)

Panda is a cybersecurity brand (antivirus software), not a type of firewall.

Q.58**NETWORKING**

"S" in HTTPS stands for _____.

- a) Safe
- b) Secure**
- c) System
- d) Server

✓ ANSWER: (B)

HTTPS = HyperText Transfer Protocol **Secure**.

Q.59**HASHING**

The process used to get an output 43205 from a given input "DCTE" is

- a) Hashing**
- b) Compression
- c) Decompression
- d) Encryption

✓ ANSWER: (A)

Hashing converts an input of any size into a fixed-length alphanumeric string.

$a = 10$, $b = 20$, $a = a + b$, $b = a - b$, $a = a - b$. The final values of a and b are:

a) $a=30$, $b=10$

b) $a=30$, $b=20$

c) $a=20$, $b=10$

d) $a=20$, $b=30$

✓ ANSWER: (C)

$a=10$, $b=20$

$a = a+b = 30$

$b = a-b = 30-20 = 10$

$a = a-b = 30-10 = 20$

$\therefore a=20$, $b=10$ (values are swapped)

04

CHAPTER FOUR

Statistics & Analytics

20 Marks

Questions 61 – 80

Q.61

QUESTIONNAIRES

Which one of the following is not essential for good questionnaires?

- a) Logically arranged
- b) Avoiding personal questions
- c) Limiting number of questions
- d) Unlimited number of questions

✓ ANSWER: (D)

Unlimited number of questions is not essential for a good questionnaire.

Q.62

FREQUENCY

Both categorical and numerical data variables can be used for

- a) Continuous variable
- b) Class intervals
- c) Distribution of frequency
- d) Frequency distribution table

✓ ANSWER: (D)

Frequency distribution table can be used for both categorical and numerical data.

Q.63

EXCEL

Match the following:

COLUMN A

- a) Mean
- b) Absolute value
- c) IQR

COLUMN B

- i) $Q_3 - Q_1$
- ii) =AVERAGE(array)
- iii) =ABS(cell)

- a) a-i, b-ii, c-iii
- b) a-ii, b-iii, c-i
- c) a-ii, b-i, c-iii
- d) a-i, b-iii, c-ii

✓ ANSWER: (B)

Mean → =AVERAGE(array), Absolute value → =ABS(cell), IQR → $Q_3 - Q_1$

_____ type of chart is used to show the relationship between numerical and categorical variable

- a) Plot
- b) Pie
- c) Bar**
- d) Line

✓ ANSWER: (C)

Bar chart shows the relationship between numerical and categorical variables.

Who developed Python language?

- a) Zim Den
- b) Guido van Rossum**
- c) Niene Stom
- d) Wick van Rossum

✓ ANSWER: (B)

Python was created by **Guido van Rossum**.

Which one of the following is not the benefit of data cleaning?

- a) Removal of errors when multiple sources of data are at play
- b) Boosts results and revenue
- c) Helps with faster decision-making and more efficient business practices
- d) Consumes more time**

✓ ANSWER: (D)

Consuming more time is not a benefit of data cleaning.

Is Python code compiled or interpreted?

- a) Python code is both compiled and interpreted
- b) Python code is neither compiled nor interpreted
- c) Python code is only compiled
- d) Python code is only interpreted

✓ ANSWER: (A)

Python code is first compiled into bytecode, then interpreted by the Python Virtual Machine (PVM).

Which one of the following is not a type of interview?

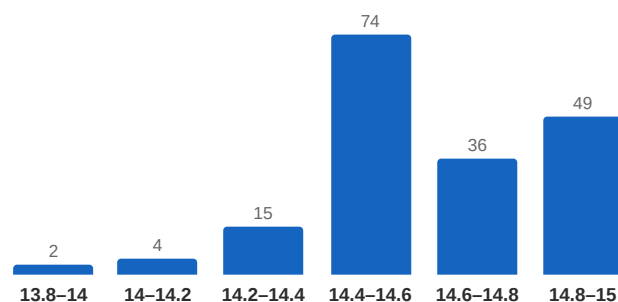
- a) Unstructured
- b) Semi-structured
- c) Audio recorder
- d) Structured

✓ ANSWER: (C)

Audio recorder is not a type of interview.

The time taken by 180 athletes to run a 110m hurdle race is tabulated. The number of athletes who completed the race in less than 14.6 seconds is:

Frequency Distribution — 110m Hurdle Race (180 Athletes)



- a) 19
- b) 95
- c) 131
- d) 180

✓ ANSWER: (B)

Athletes who completed in < 14.6 s:
 $= 2 + 4 + 15 + 74 = 95$

Q.70**RANGE**

The higher value of the range represents:

- ☒ a) greater spread of the Data
- ☐ b) lesser spread of the Data
- ☐ c) equal to range of the Data
- ☐ d) average of the Data

✓ ANSWER: (A)

A higher range value indicates **greater spread** of data.

Q.71**CENTRAL TENDENCY**

The middle number in a given data set is called:

- ☐ a) Mean
- ☐ b) Mode
- ☐ c) Frequency
- ☒ d) Median

✓ ANSWER: (D)

Median is the middle-most value in arranged data.

Q.72**DATA TYPES**

_____ is the example of qualitative data.

- ☐ a) Temperature
- ☒ b) Ability of students
- ☐ c) Prices
- ☐ d) Area

✓ ANSWER: (B)

Ability of students is not numerically measurable — it is qualitative data.

Q.73**DATA COLLECTION**

Direct personal observation is a method of collection of:

- ☐ a) Secondary Data
- ☒ b) Primary Data
- ☐ c) Mixed Data
- ☐ d) Unpublished Data

✓ ANSWER: (B)

Direct personal observation is a method of collecting **primary data**.

Q.74

EXCEL

The in-built function to find the mean of a given data in Excel is:

- a) `=AVERAGE()` b) `=MEAN()`
c) `=MEDIAN()` d) `=MODE()`

✓ ANSWER: (A)

Excel formula for mean is `=AVERAGE()`

Q.75

FREQUENCY

The formula for Relative Frequency is:

- a) Frequency × No. of observations b) $\frac{\text{No. of observations}}{\text{Frequency}}$
c) $\frac{\text{Frequency}}{\text{No. of observations}}$ d) No. of observations + Frequency

✓ ANSWER: (C)

Relative frequency = $\frac{f}{n}$

Q.76

PYTHON

In Python, "str" is a:

- a) Numeric type b) Binary type
c) Text type d) Sequence type

✓ ANSWER: (C)

`str` is the Text type in Python.

Q.77

PYTHON

The results of Python program get displayed in

- a) IDLE 3.1.9 Window b) IELD shell 3.9.1 Window
c) IDLE shell 3.9.1 Window d) ILDE shell 3.1.9 Window

✓ ANSWER: (C)

`IDLE` (Integrated Development and Learning Environment) shell 3.9.1 Window.

The process of Data cleaning is

- a) Removing virus
- b) Deduplication
- c) Survey
- d) Google form

✓ ANSWER: (B)

Data cleaning involves removing duplicate data, called **deduplication**.

Data collected for the first time is called:

- a) Published Data
- b) Mixed Data
- c) Primary Data
- d) Secondary Data

✓ ANSWER: (C)

Data collected for the first time directly from the field by an investigator is called **primary data**.

The Excel formula for standard deviation of a sample is:

- a) STDEV.P
- b) SDTEV.S
- c) STDEV.S
- d) SDTEV.P

✓ ANSWER: (C)

Excel formula for sample standard deviation is **=STDEV.S**

05

CHAPTER FIVE

Fundamentals of Electrical & Electronics Engineering

20 Marks

Questions 81 – 100

Q.81

SAFETY

CPR stands for

- a) Cardio Permanent Respiration b) Cardio Pulmonary Respiration
c) Cardio Pulmonary Resuscitation d) Cardio Pulmonary Retrieval

✓ ANSWER: (C)

CPR stands for **Cardio Pulmonary Resuscitation**.

Q.82

CIRCUITS

Two 5Ω resistors are connected in parallel and this combination is connected in series with a 2.5Ω resistor. The effective resistance is:

- a) 0.5Ω b) 5Ω
c) 2.5Ω d) 7.5Ω

✓ ANSWER: (B)

$$R_p = \frac{5 \times 5}{5 + 5} = \frac{25}{10} = 2.5\Omega$$

$$\text{Total} = R_p + 2.5 = 2.5 + 2.5 = 5\Omega$$

Q.83

OHM'S LAW

The relationship between Voltage (V), Current (I) and Resistance (R) is:

- a) $I = VR$ b) $R = VI$
c) $V = IR$ d) $V = I/R$

✓ ANSWER: (C)

Ohm's Law: **$V = IR$**

Q.84**PROTECTION**

ELCB stands for:

- a) Earth Line Circuit Breaker
- b) Electric Load Circuit Breaker
- c) Earth Leakage Circuit Breaker**
- d) Electric Line Circuit Breaker

✓ ANSWER: (C)

ELCB stands for **Earth Leakage Circuit Breaker**.

Q.85**FUSE**

Good fuse wire must have:

- a) Low melting point**
- b) High melting point
- c) High resistivity
- d) High melting point and high resistivity

✓ ANSWER: (A)

A good fuse wire must have **low melting point** to break the circuit quickly on overload.

Q.86**MOTORS**

A 10 HP submersible pump set is used in a borewell for irrigation purpose. It normally uses _____ type of motor.

- a) Single-phase induction motor
- b) Three-phase induction motor**
- c) DC motor
- d) FHP motor

✓ ANSWER: (B)

A **three-phase induction motor** is used in borewells for irrigation with 10 HP pump sets.

Q.87**BATTERIES**

SMF in batteries stands for:

- a) Sealed Moulded - Free
- b) Semiconductor Maintenance - Free
- c) Sealed Maintenance - Free**
- d) Solar Maintenance - Free

✓ ANSWER: (C)

SMF stands for **Sealed Maintenance Free** batteries.

Q.88**SENSORS**

The typical use of LVDT is to sense

- a) Light intensity
- b) Humidity
- c) Pressure**
- d) Temperature

✓ ANSWER: (C)

LVDT sensor is used to sense displacement, force, and **pressure**.

Q.89**ELECTRONICS**

A p-n junction diode is a

- a) Bidirectional switch
- b) Unidirectional switch**
- c) Bidirectional Controlled Switch
- d) Unidirectional Controlled Switch

✓ ANSWER: (B)

A P-N junction diode is a **unidirectional switch** — it allows current flow only from anode to cathode.

Q.90**RECTIFIERS**

The purpose of C Filter in a power supply is to

- a) Degrade regulation
- b) Reduce ripple**
- c) Reduce gain
- d) Reduce the chance of electric shock

✓ ANSWER: (B)

The C filter is used to decrease unwanted **ripple voltage**. It is connected in parallel with the load.

Q.91**BATTERIES**

Lead acid cell is an example of

- a) Secondary cell**
- b) Solar cell
- c) Primary cell
- d) Fuel cell

✓ ANSWER: (A)

A **secondary cell** is designed to be recharged and reused. Examples: Lead-Acid, Nickel Cadmium, Lithium ion.

Time taken by the alternating quantity to complete one cycle is called

- a) Frequency
- b) Time period
- c) Amplitude
- d) Wavelength

✓ ANSWER: (B)

The time taken to complete one cycle is called **time period**.

What does "PLC" stand for in the context of Industrial Automation?

- a) Programmable Logic Circuit
- b) Process Logic Controller
- c) Programmable Logic Controller
- d) Processor Logic Controller

✓ ANSWER: (C)

PLC stands for **Programmable Logic Controller** used for industrial automation.

Maximum value attained by the alternating quantity either in positive half-cycle or negative half-cycle is called

- a) Frequency
- b) Altitude
- c) Amplitude
- d) Wavelength

✓ ANSWER: (C)

Amplitude is the maximum value attained by an alternating quantity in either half cycle.

Which of the following statement is correct?

- a) In series circuit the current through each resistor is different
- b) In parallel circuit the voltage across each resistor is same**
- c) In parallel circuit the equivalent resistance is sum of individual resistances
- d) In series circuit the reciprocal of equivalent resistance is sum of reciprocal of individual resistances

✓ ANSWER: (B)

In a parallel circuit, the **voltage remains constant** across each resistor.

The cell which cannot be recharged and reused is

- a) Secondary cell
- b) Battery
- c) Primary cell**
- d) Simple cell

✓ ANSWER: (C)

A **primary cell** is designed to be used once and discarded; it cannot be recharged.

Specific gravity of a fully charged lead acid battery will be in the range of

- a) 1000 to 1210
- b) 1100 to 1310
- c) 1150 to 1310
- d) 1230 to 1300**

✓ ANSWER: (D)

Specific gravity of a fully charged lead acid battery is in the range of **1230 to 1300**.

The best sensor for sensing vibration is

- a) Thermistor
- b) Thermocouple
- c) Piezoelectric crystal**
- d) Hygrometer

✓ ANSWER: (C)

The **piezoelectric crystal** sensor is used for sensing vibration.

During charging of batteries

- a) Chemical → Mechanical energy
- b) Mechanical → Chemical energy
- c) Electrical → Chemical energy**
- d) Chemical → Electrical energy

✓ ANSWER: (C)

During charging, **electrical energy is converted to chemical energy**.

When an electric stove is connected across 230 volts, 50 Hz AC supply for one hour and takes a current of 10 amperes, the energy consumed is:

- a) 230 units
- b) 2.3 units**
- c) 23 units
- d) 0.23 units

✓ ANSWER: (B)

$$\begin{aligned}\text{Energy} &= V \times I \times t = 230 \times 10 \times 1 = 2300 \text{ Wh} \\ &= \frac{2300}{1000} = 2.3 \text{ units (kWh)}\end{aligned}$$



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